

Ex.No.8 Use Steg-Expose to detect hidden data in images

Description

StegExpose is a tool used to detect hidden data in images using steganography analysis. It works by evaluating the statistical properties of an image to estimate whether any hidden data is embedded. Below is a step-by-step guide on how to use StegExpose to detect hidden data in images:

Prerequisites:

1. **Java Runtime Environment (JRE):** StegExpose is a Java-based tool, so ensure you have JRE installed.
2. **StegExpose Tool:** Download StegExpose from the [official GitHub repository](#).

Step-by-Step Process:

1. Download and Set Up StegExpose:

- **Download the tool:** Go to [StegExpose GitHub](#) and download the .jar file.
- **Install Java:** Ensure that Java is installed on your machine. You can download it from the official site.
- **Prepare environment:** Place the .jar file in a folder where you plan to work on the image files.

2. Select Images for Analysis:

- Collect the images you suspect might have hidden data. StegExpose supports formats such as .png, .jpg, .bmp, etc.

3. Open Command Line or Terminal:

- Navigate to the folder where you have saved the StegExpose.jar file.
- Open your **Command Prompt** (Windows) or **Terminal** (Linux/macOS).

4. Run StegExpose on an Image:

- Use the following command to check a single image for hidden data:

```
bash
```

```
java -jar StegExpose.jar <image_file_path>
```

Example:

```
bash
```

```
java -jar StegExpose.jar test_image.png
```

- Replace <image_file_path> with the actual path of the image you want to check.

3. Analyze the Output:

- After running the tool, it will analyze the image and provide a score.
- **Output Explanation:**

- StegExpose calculates a "suspect" score (between 0 and 1). The higher the score, the more likely that hidden data is present.
- **Threshold values** (suggested by the tool):
 - **Less than 0.2:** Image is clean (no hidden data).
 - **0.2 - 0.3:** Possibly some hidden data.
 - **Above 0.3:** Likely that steganography is present.

6. Batch Analysis (Multiple Images):

- If you have multiple images to check, you can analyze them in bulk by specifying a folder. For example:

```
bash
```

```
java -jar StegExpose.jar <folder_path>
```

- Replace <folder_path> with the path to the folder containing the images.

7. Advanced Options (Optional):

- StegExpose provides additional parameters, such as adjusting the sensitivity of detection and output verbosity. Use --help to see the list of options:

```
bash
```

```
java -jar StegExpose.jar --help
```

8. Review the Results:

- For each image analyzed, review the scores and determine whether you need to investigate further.

Example:

```
bash
```

```
java -jar StegExpose.jar suspect_image.png
```

```
Output:
```

```
makefile
```

```
Analyzing suspect_image.png...
```

```
Result: 0.4
```

```
Steganography likely present
```

This output indicates that hidden data is likely present in the image suspect_image.png.

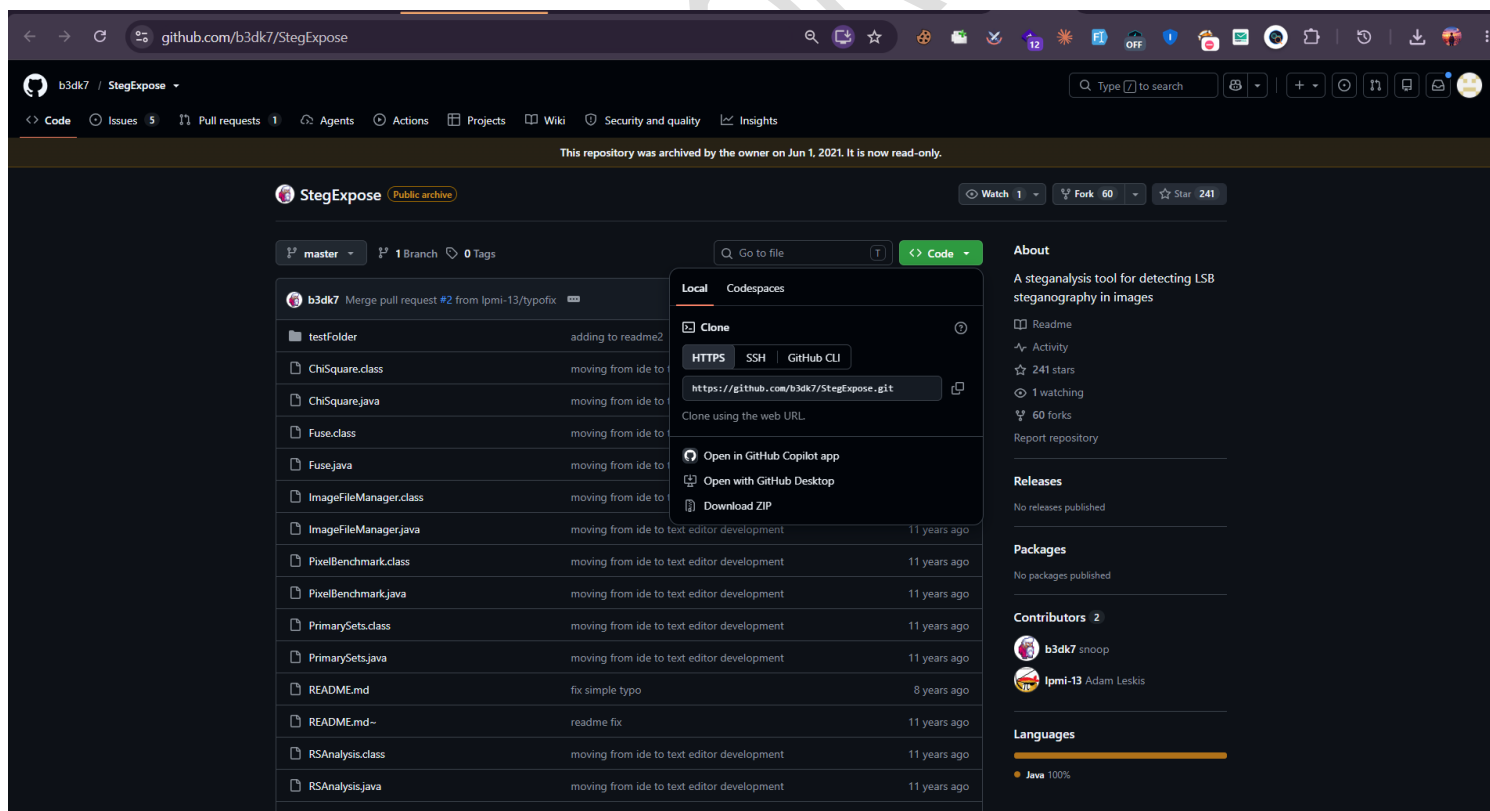
Step 1: Java Runtime Environment

```
Command Prompt
Microsoft Windows [Version 10.0.26200.9457]
(c) Microsoft Corporation. All rights reserved.

C:\Users\saiha>java -version
java version "1.8.0_503"
Java(TM) SE Runtime Environment (build 1.8.0_503-b01)
Java HotSpot(TM) 64-Bit Server VM (build 25.503-b01, mixed mode)

C:\Users\saiha>
```

Step 2: Downloading StegExpose Tool



Step 3: Test on single image file (Output)

```
C:\Users\saiha\Downloads\StegExpose-master\StegExpose-master>java -jar StegExpose.jar img
stego_6666458261_e455d262b5_z.png is suspicious. Approximate amount of hidden data is 114785 bytes.

C:\Users\saiha\Downloads\StegExpose-master\StegExpose-master>
```

Step 4: Test on the Directory (Output)

```
C:\Users\saiha\Downloads\StegExpose-master\StegExpose-master>java -jar StegExpose.jar testFolder
stego_6666458261_e455d262b5_z.png is suspicious. Approximate amount of hidden data is 114785 bytes.
stego_6672108499_85c582a7f9.png is suspicious. Approximate amount of hidden data is 137047 bytes.
stego_6672542201_532f70bffe.png is suspicious. Approximate amount of hidden data is 67141 bytes.

C:\Users\saiha\Downloads\StegExpose-master\StegExpose-master>
```

Step 5: Steganography Scores (Output)

```
C:\Users\saiha\Downloads\StegExpose-master\StegExpose-master>java -jar StegExpose.jar testFolder default default result.csv
C:\Users\saiha\Downloads\StegExpose-master\StegExpose-master>type result.csv

File name,Above stego threshold?,Secret message size in bytes (ignore for clean files),Primary Sets,Chi Square,Sample Pairs,RS analysis,Fusion (mean)
clean_7221186570_82a10bc040_z.png,false,13416,0.11329240178885477,0.1423569884449125,null,0.048735256189453034,0.10146154880774011
clean_7222762386_2688184c8e.png,false,7391,NaN,3.741100013179327E-11,0.24006602558881854,0.25861310226721124,0.16622637596448026
clean_7228374452_8f743dc3d4_z.png,false,2449,0.031137468869920987,0.0073545822757012215,null,0.01607070559017578,0.01818758557859933
clean_7228718722_1cf25dff3e_z.png,false,14194,0.116140905612938,0.0021023440017611224,null,0.09988112328129448,0.07270812429866454
clean_7231623538_f51db2a35a_z.png,false,8198,0.050937672916859474,0.018913031823677696,null,0.06165268087585292,0.043834461872130026
clean_7232193260_c2fd8c0f25_z.png,false,22569,0.2573883057340656,0.002651049049337065,0.09692103046233898,0.10675110692744194,0.11592787304593088
clean_7232202430_d38b6d6986_z.png,false,25126,NaN,0.009698055773106866,0.18017057666719039,0.21013951225694777,0.133336048232415
clean_7232206610_b8cfdded120_z.png,false,10190,0.028480431325644658,0.03596650763921852,0.080529823771301,0.08440531345457021,0.05734551904768359
clean_7232220662_3d42c69109_z.png,false,18000,NaN,0.01252686945037774,0.13287065993498356,0.1474074177378486,0.09760164904106998
clean_7234473324_bb8a82b5bd.png,false,1403,0.01653992591872635,0.0020576187477316554,null,0.036616633877419855,0.01840472618129262
clean_7235558256_3099066753.png,false,21450,0.17546102698982005,0.1769494348209659,null,0.20167053078214375,0.18469366419764324
clean_7235972310_8c25258da5.png,false,10882,0.18496458785274303,0.005156703935488528,null,0.0972843399655833,0.09580187725127161
clean_7237798856_86870c8823_z.png,false,3654,0.07330206436492333,0.569594619348558E-4,null,0.06965819797476018,0.04793907393387279
stego_6666458261_e455d262b5_z.png,true,114785,NaN,0.06767021195235032,0.7292902212895038,0.7380125186839619,0.5116576506419387
stego_6672108499_85c582a7f9.png,true,137047,NaN,0.9999905976238456,null,0.8698770301271956,0.9249338138755205
stego_6672542201_532f70bffe.png,true,67141,NaN,0.09371291967741853,0.7812415352997496,0.7680715936124942,0.5476753495298875

C:\Users\saiha\Downloads\StegExpose-master\StegExpose-master>
```